

Principal Component Analysis

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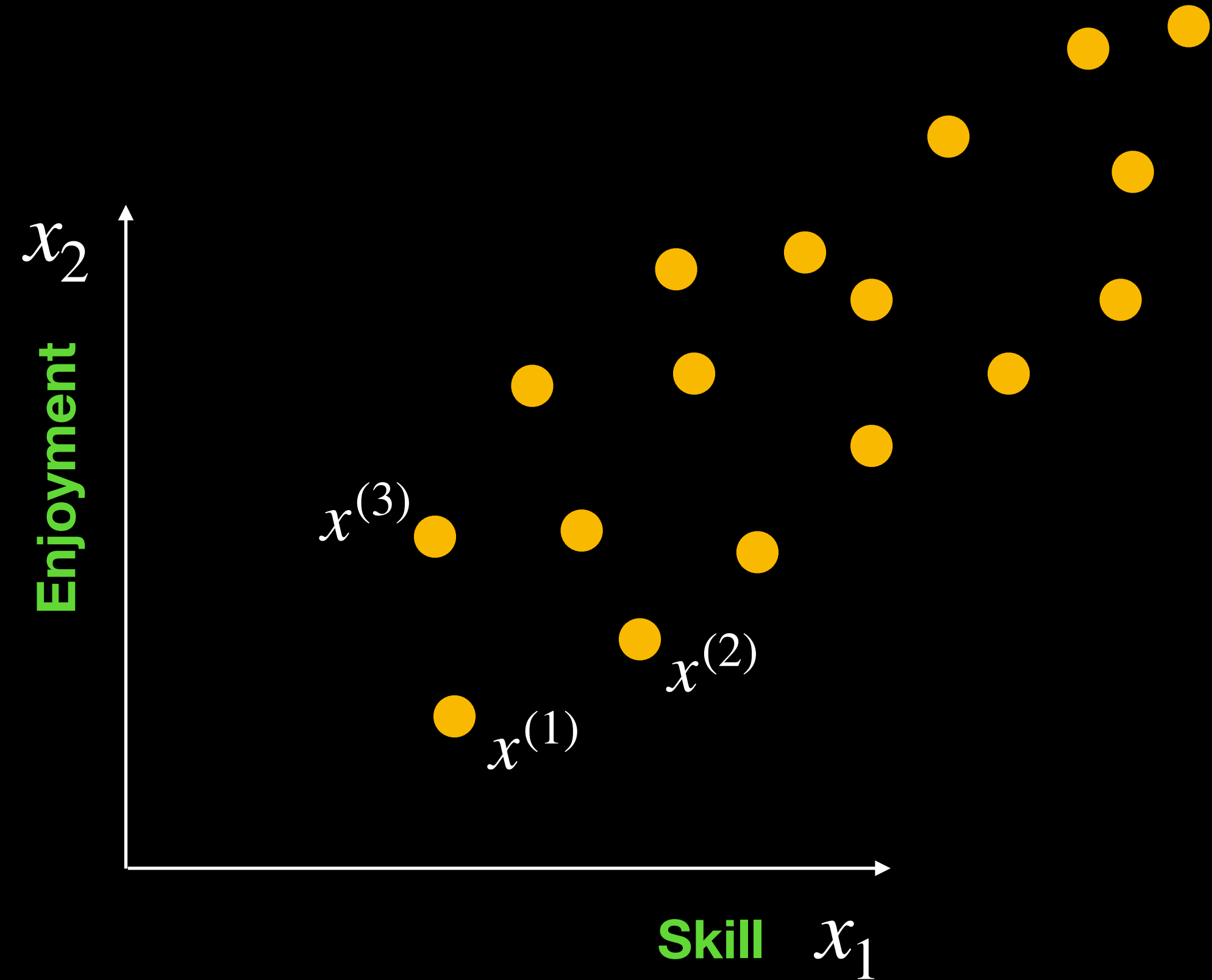
Applications

- Image Compression: Represent images (or patches) in a lower-dimensional basis (keeping dominant principal components)
- Visualization of High-Dimensional Data: Project e.g. gene expression data, text embeddings, customer features into 2–3 dims for human interpretation
- Population Genetics: Identify population structure, genetic ancestry, subpopulation clusters
- Finance: Reduce dimensionality of many stock returns, risk factors, interest rates, or yield curves
- Manufacturing & Industrial Process Monitoring: Monitor multivariate sensor data over time (temperature, pressure, vibration, etc.)
- Healthcare: Reduce gene expression, medical features, or patient record dimensions
- Marketing: From many features (demographics, purchase behavior), find fewer latent “preference axes”
- Background Modeling in Video: Reduce video into temporal and spatial modes
- Physical modeling: Find main spatial and temporal models of a continuous fluid/material

Principal Component Analysis

Dataset

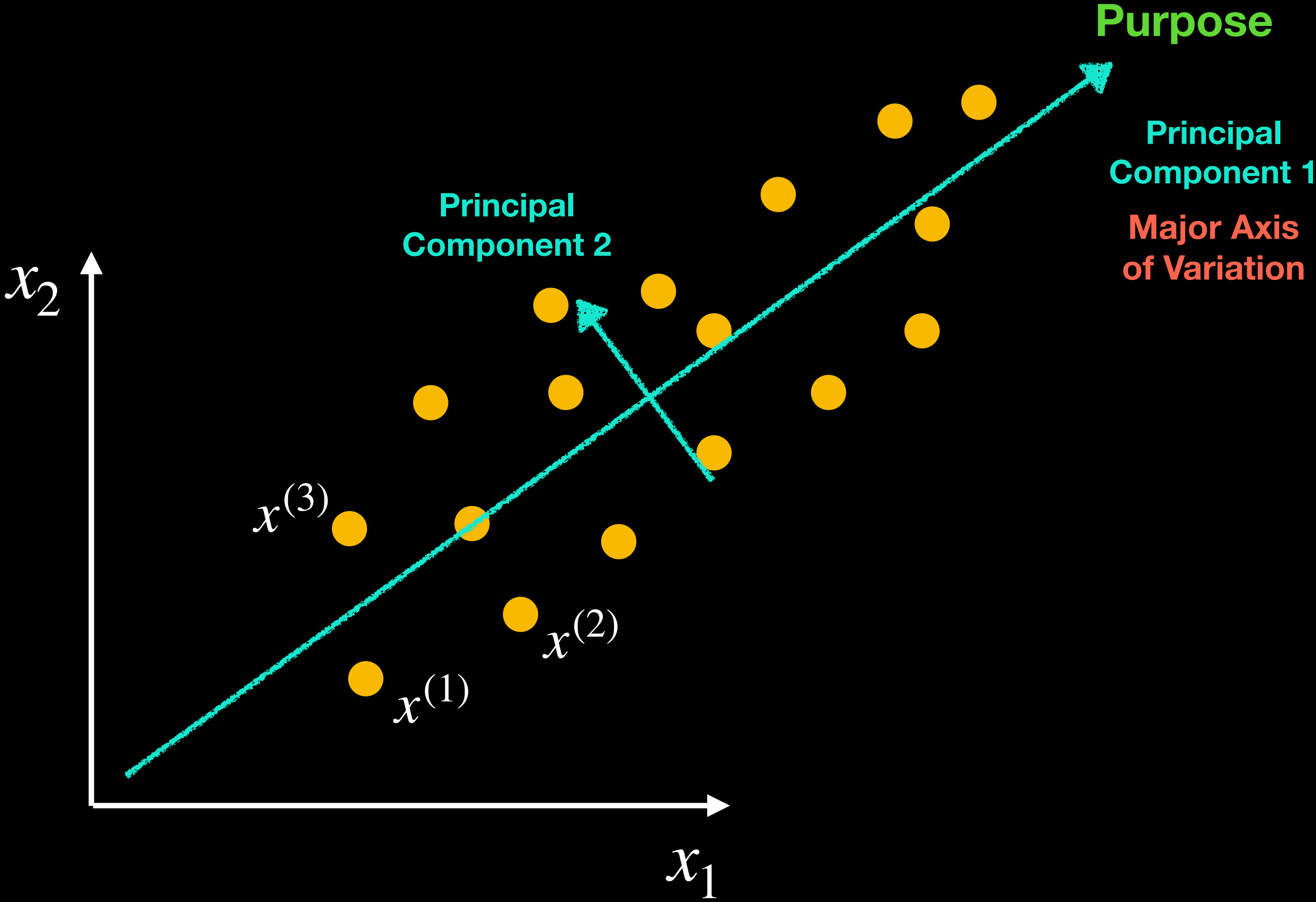
x_1	x_2
1.2	1.2
3.2	5.4
4.3	6.4
3.2	5.4
...	...



Principal Component Analysis

Dataset

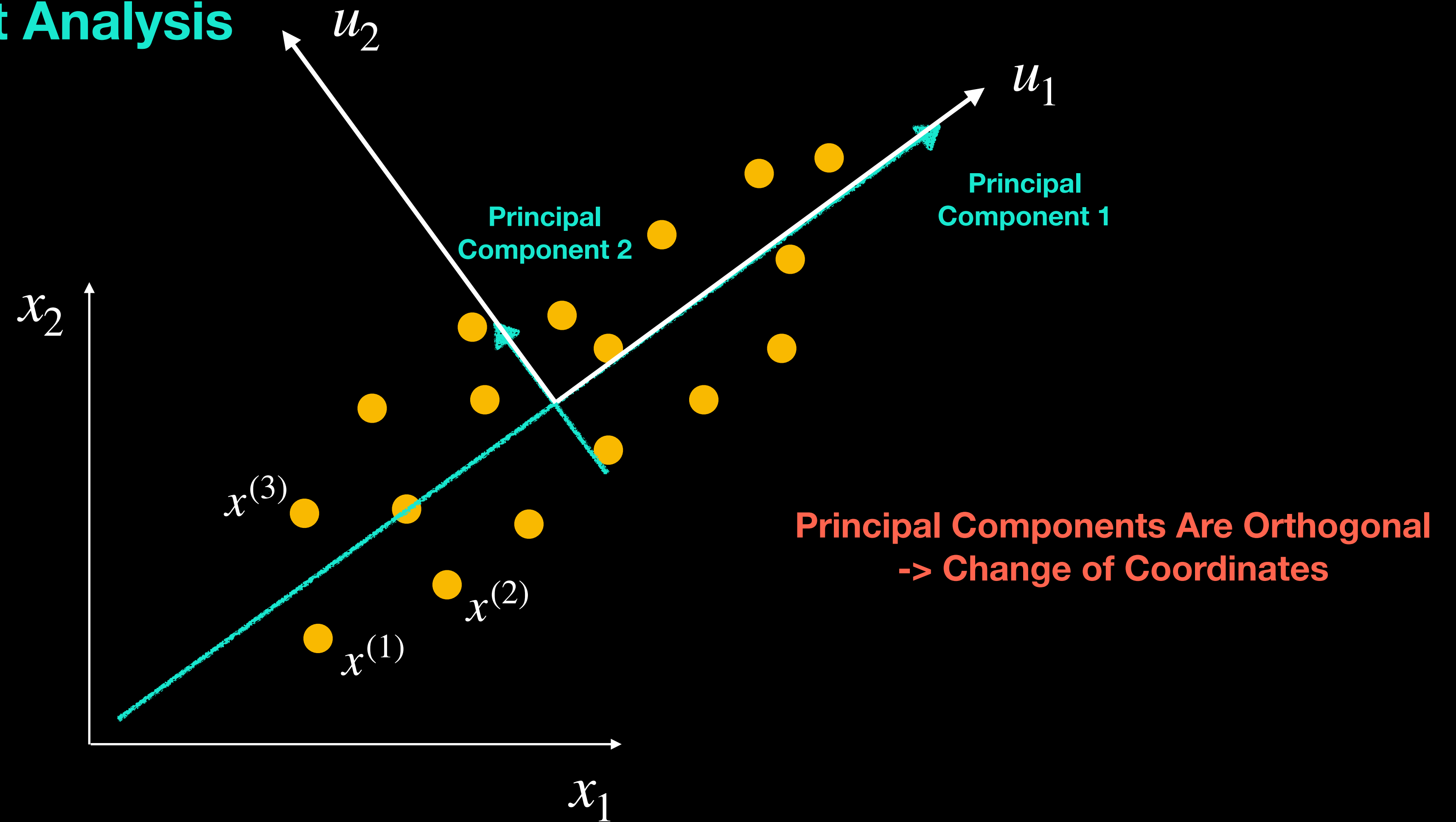
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Principal Component Analysis

Dataset

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Principal Component Analysis

Mean

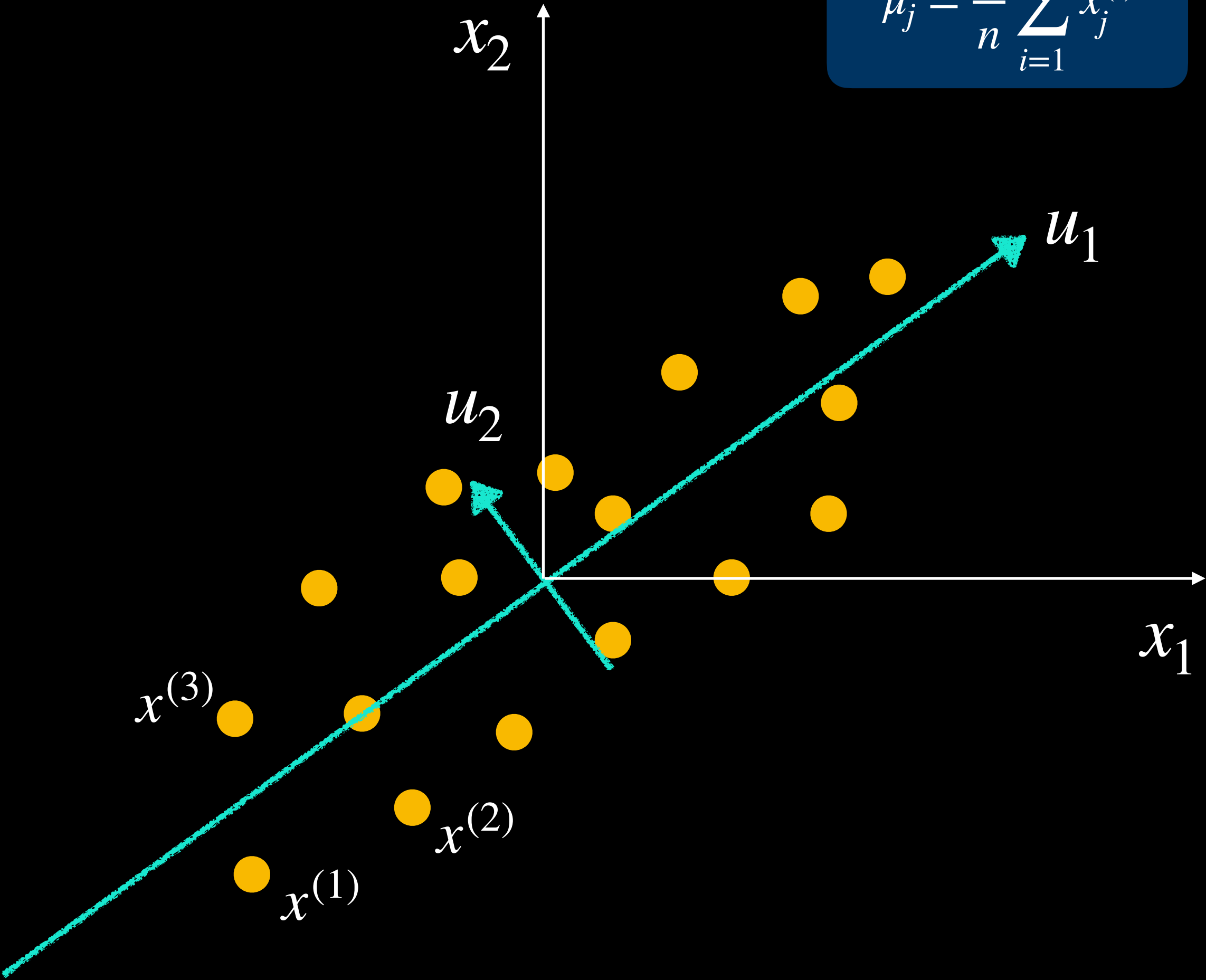
$$\mu_j = \frac{1}{n} \sum_{i=1}^n x_j^{(i)}$$

Center the data

$$x_j^{(i)} \leftarrow x_j^{(i)} - \mu_j$$

Dataset

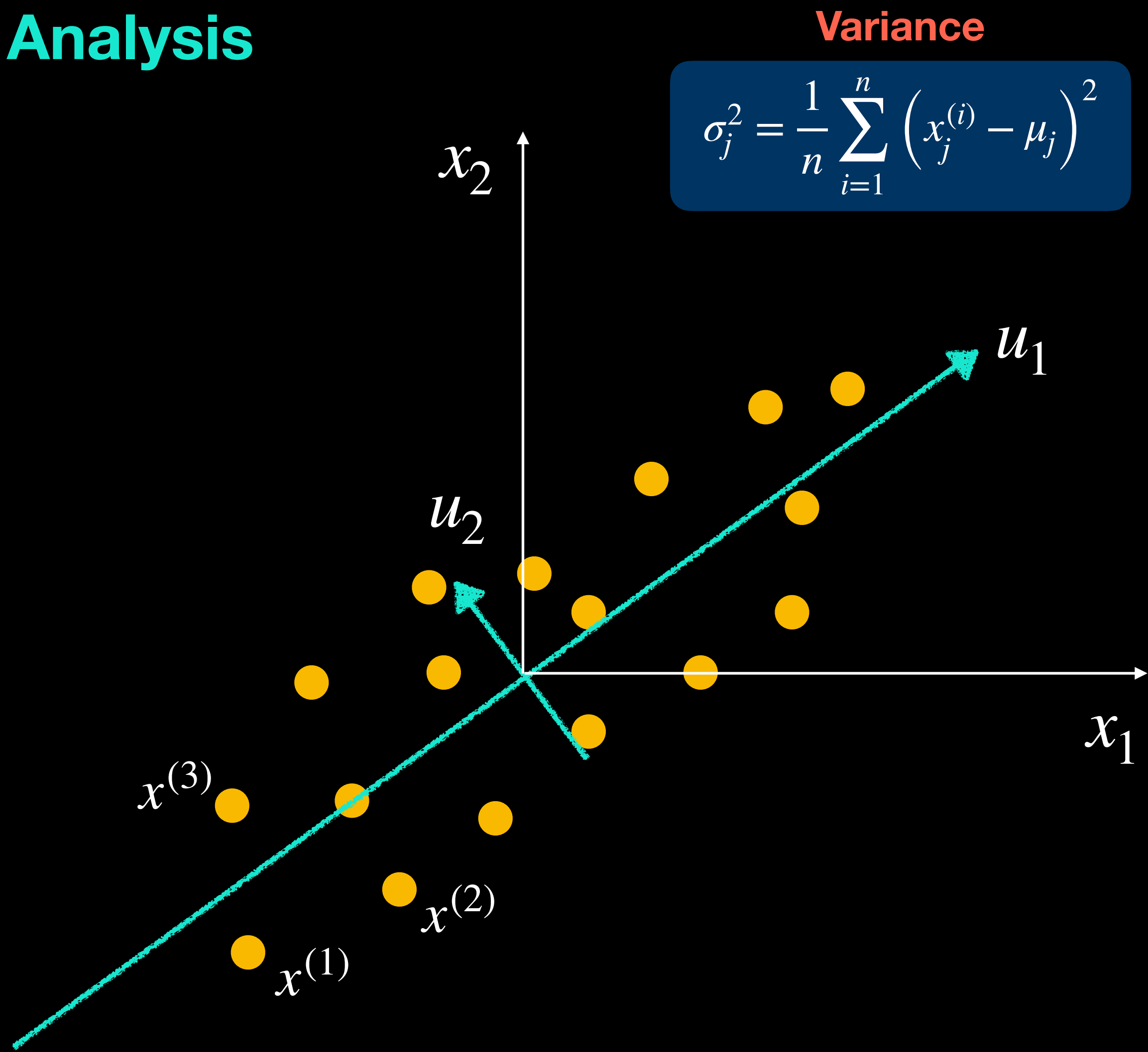
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Principal Component Analysis

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Variance

$$\sigma_j^2 = \frac{1}{n} \sum_{i=1}^n \left(x_j^{(i)} - \mu_j \right)^2$$

Normalize the data

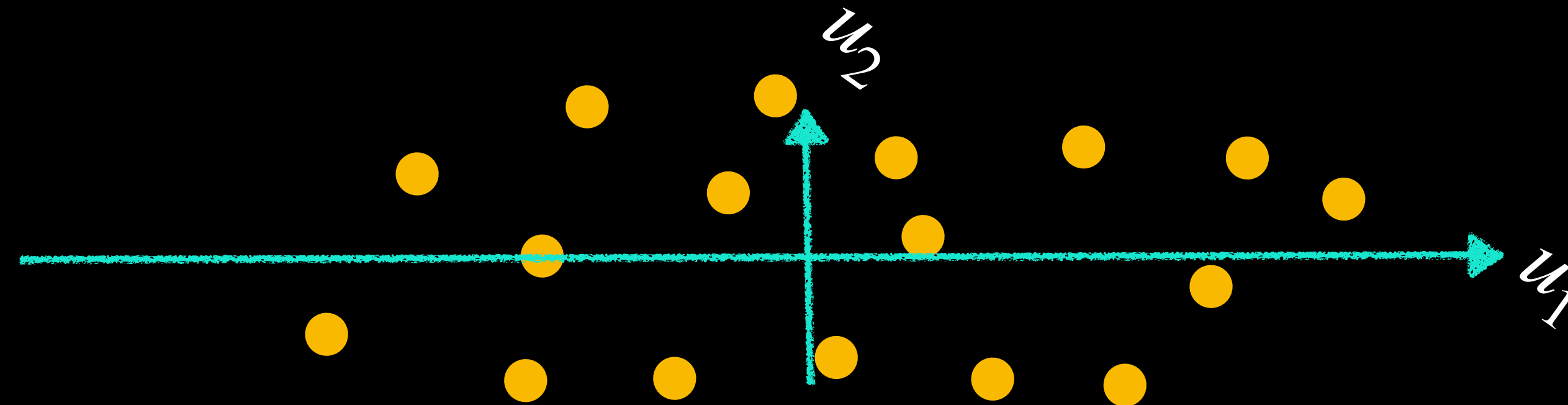
$$x_j^{(i)} \leftarrow x_j^{(i)} / \sigma_j$$

Principal Component Analysis

Data can be **projected**
On axis of highest variation: u_1

Dataset

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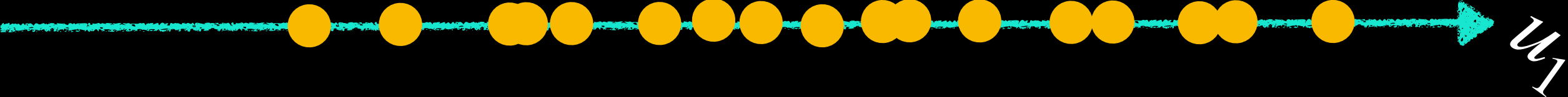


Principal Component Analysis

Data can be **projected**
On axis of highest variation: u_1

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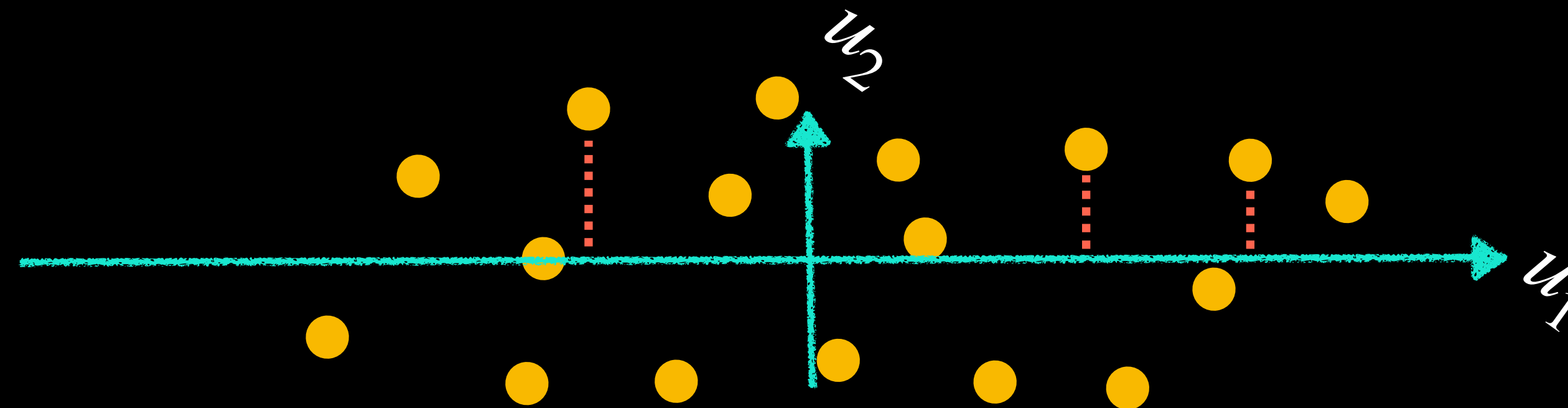


Principal Component Analysis

Error can be computed from
distances in direction of u_2

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How do we find u_1 and u_2 ?

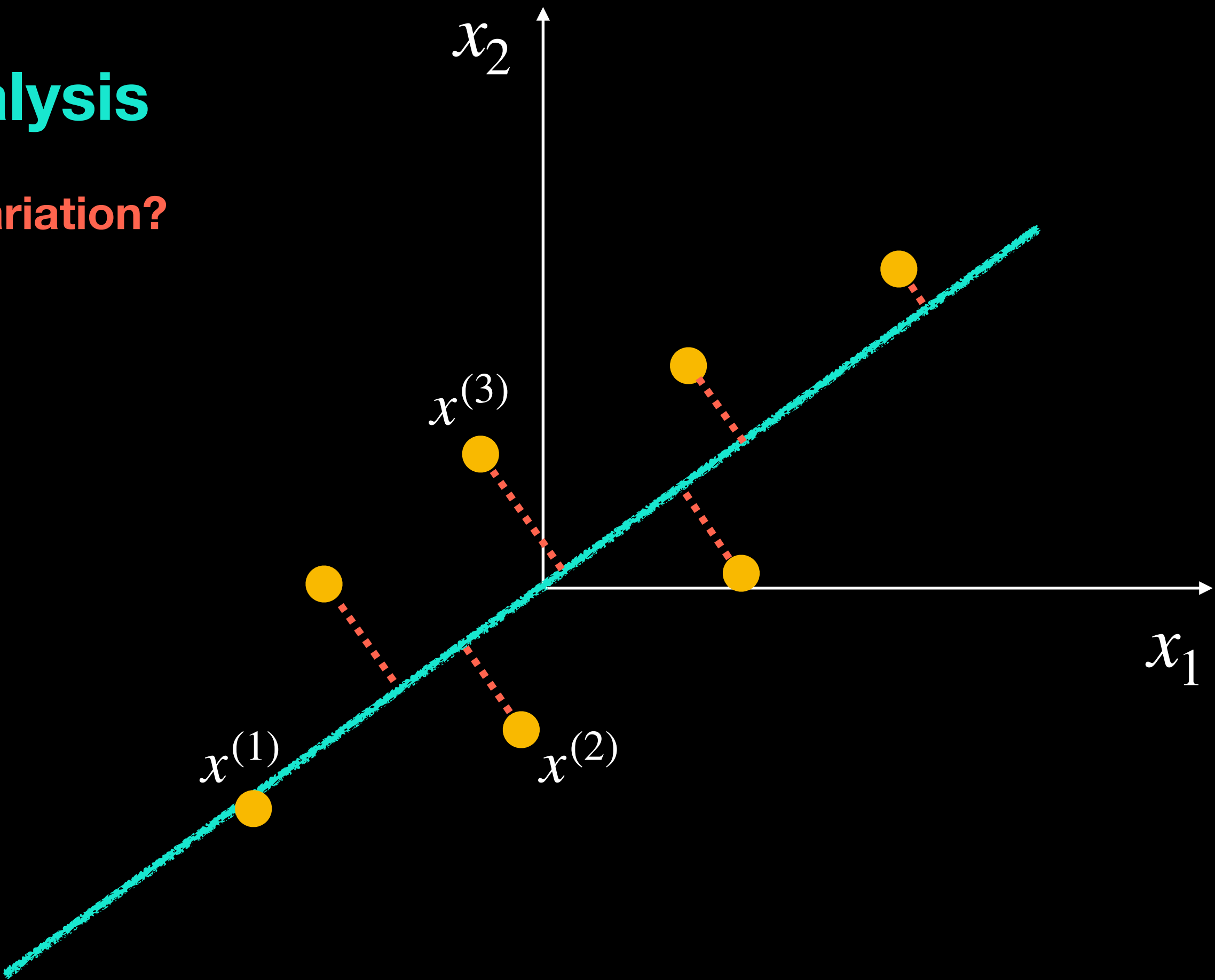
```
U, S, Vt = np.linalg.svd(data_centered)
```

Principal Component Analysis

What's the Major Axis (Direction) of Variation?

Dataset

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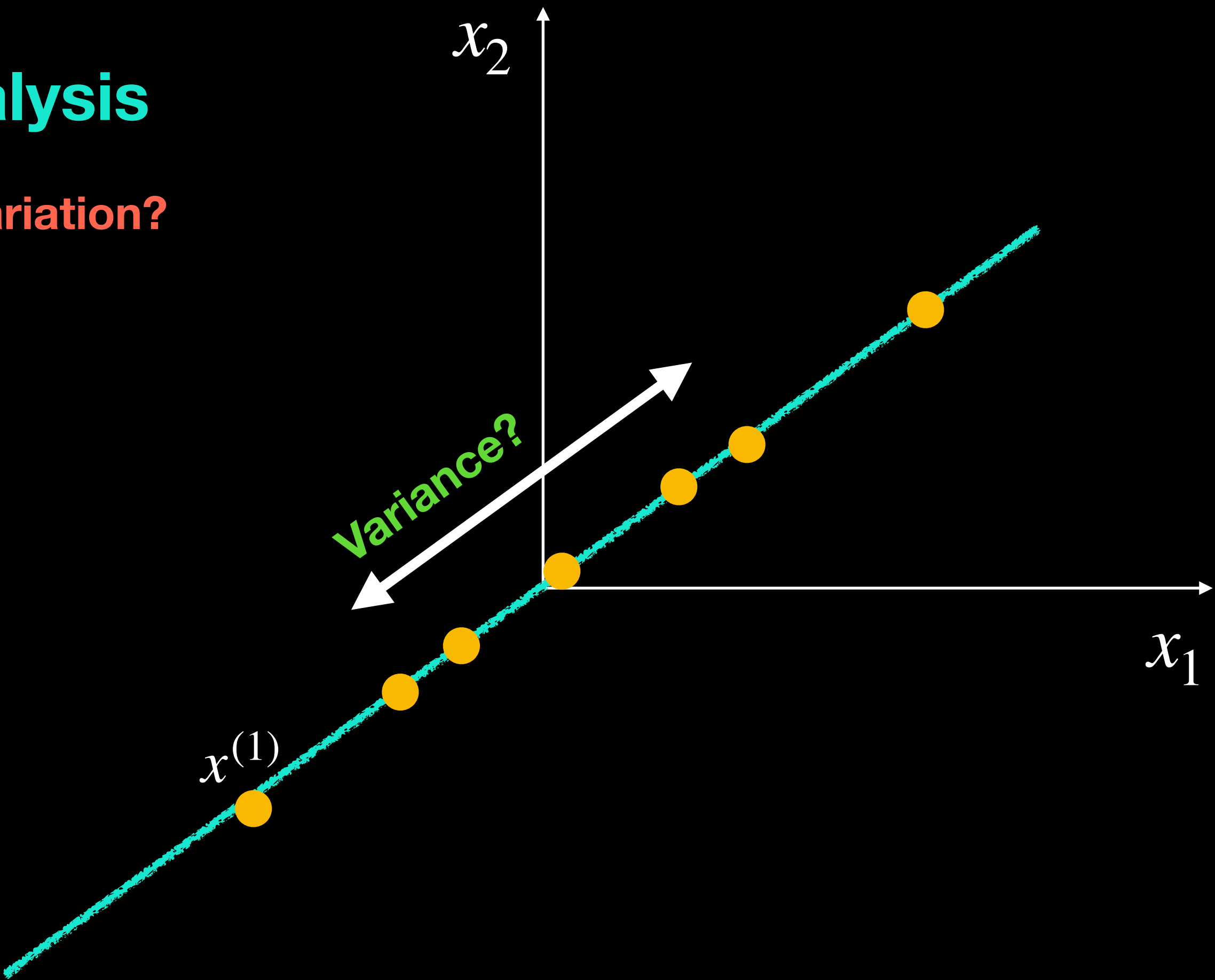


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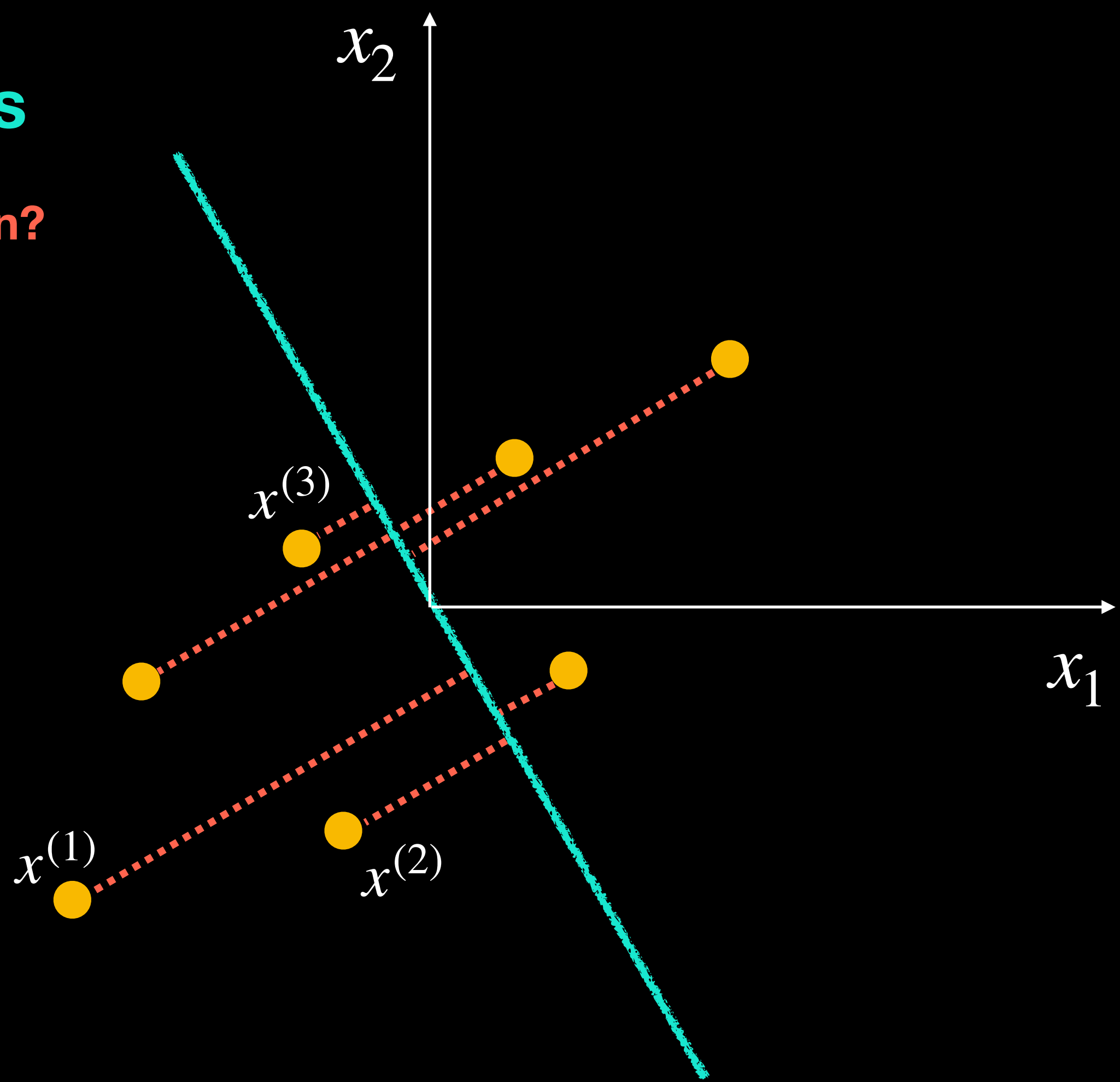


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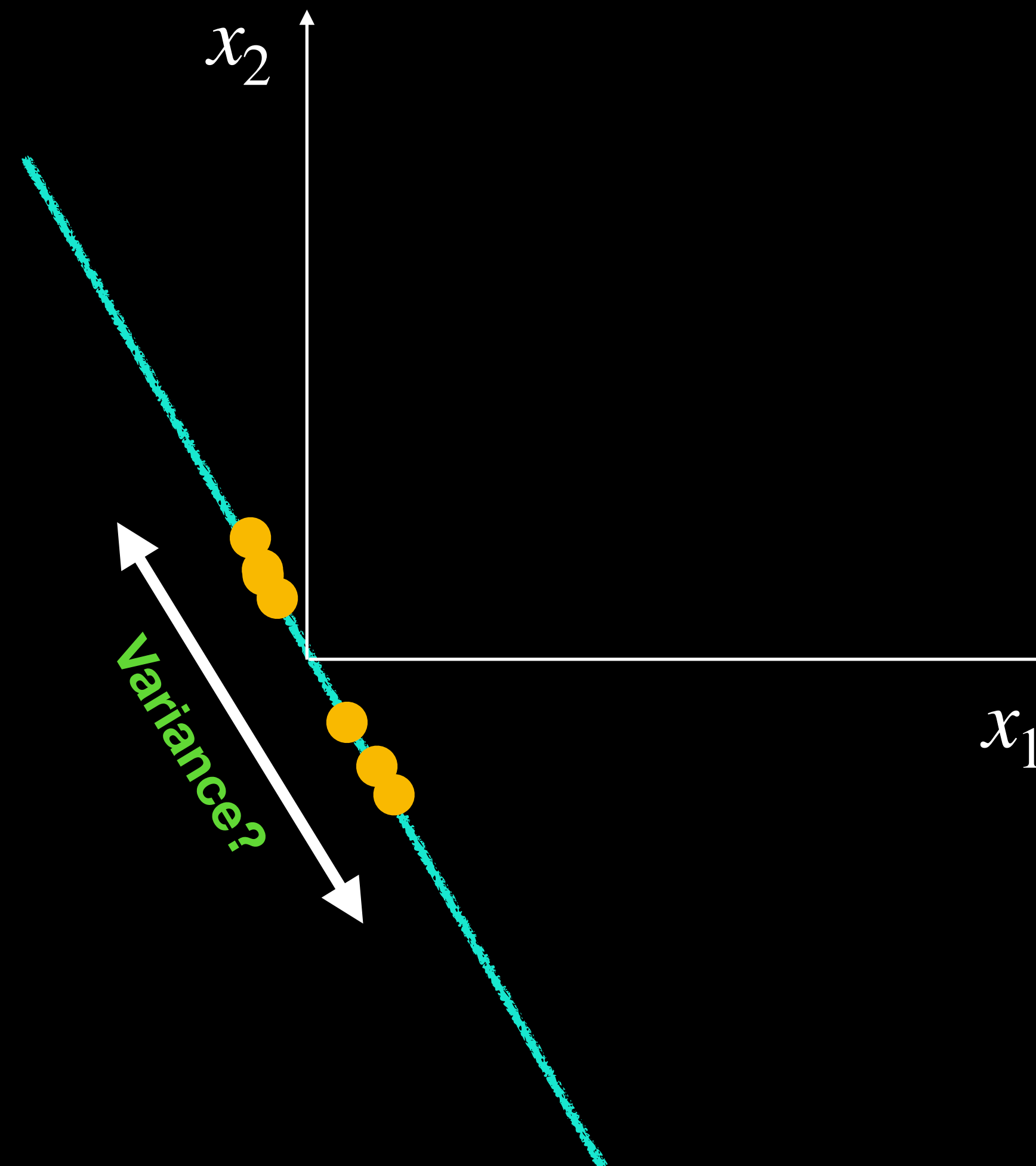


Principal Component Analysis

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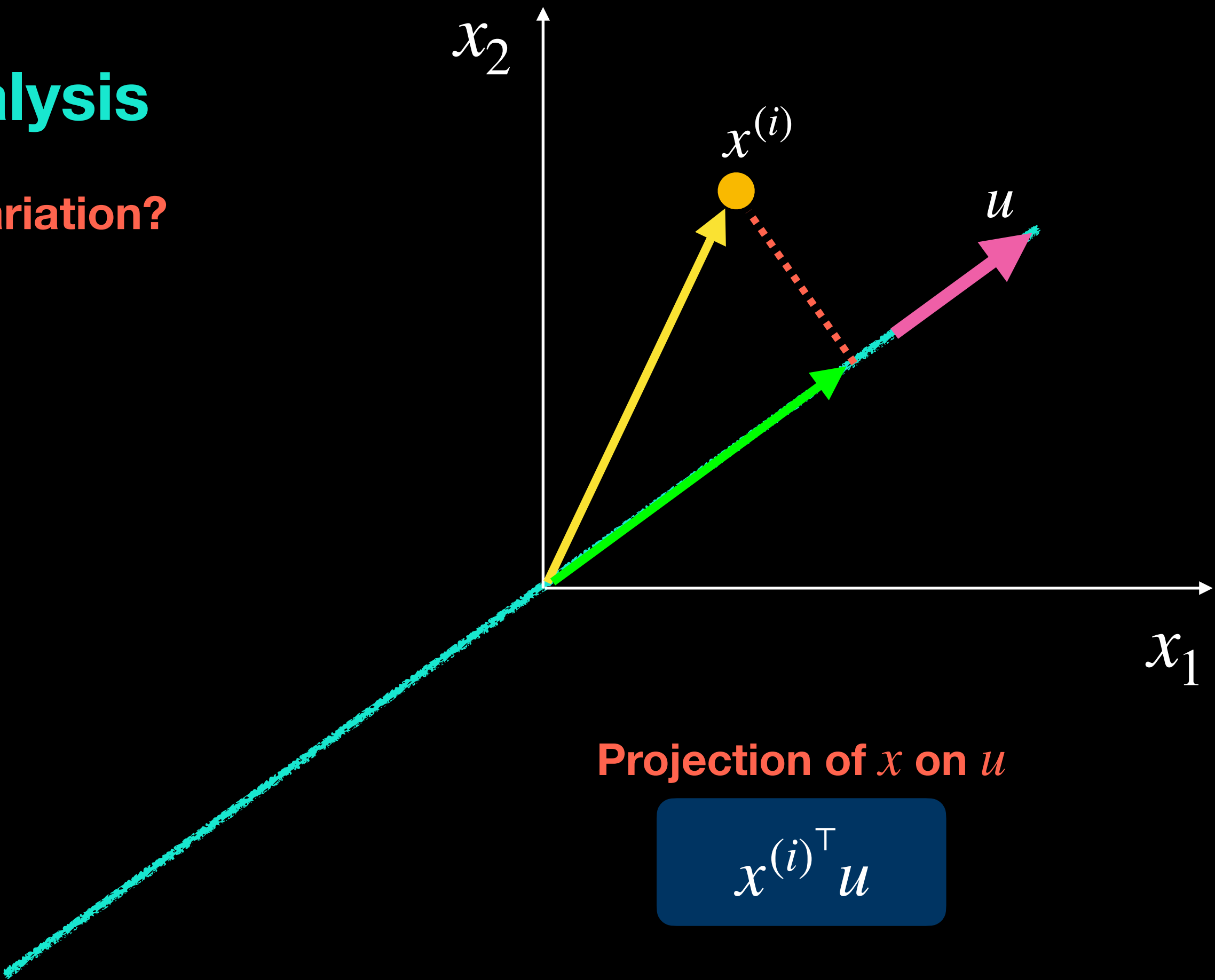


Principal Component Analysis

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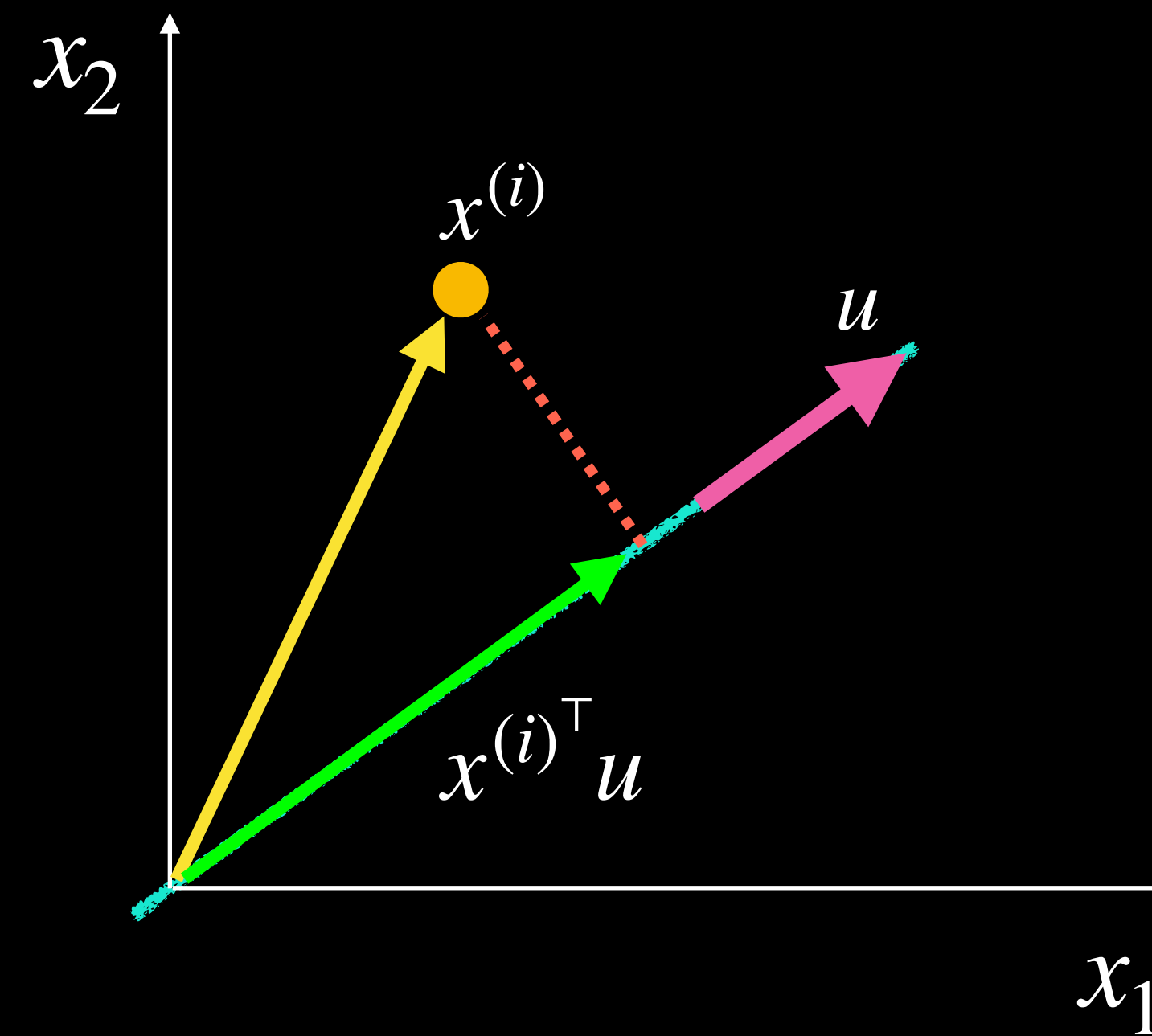
Principal Component Analysis

Maximize Variance of Projections

$$\begin{aligned} \text{Maximize} \quad & \frac{1}{n} \sum_{i=1}^n \left(x^{(i)\top} u \right)^2 \\ \text{s.t. } \|u\|_2 &= 1 \\ &= \frac{1}{n} \sum_{i=1}^n u^T x^{(i)} x^{(i)T} u \\ &= u^T \left(\frac{1}{n} \sum_{i=1}^n x^{(i)} x^{(i)T} \right) u. \end{aligned}$$

**Covariance
Matrix**

$$\max_{\|u\|=1} u^T \Sigma u$$



**Lagrange
Multipliers**

$$L(u, \lambda) = u^T \Sigma u - \lambda(u^T u - 1)$$

$$\frac{\partial L}{\partial u} = 2\Sigma u - 2\lambda u = 0$$

$$\Sigma u = \lambda u$$

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Principal Component Analysis

To Maximize Variance of Projections

Solve

$$\Sigma u = \lambda u$$

Eigenvalue problem

$$x \in \mathbb{R}^d$$
$$\Sigma \in \mathbb{R}^{d \times d}$$

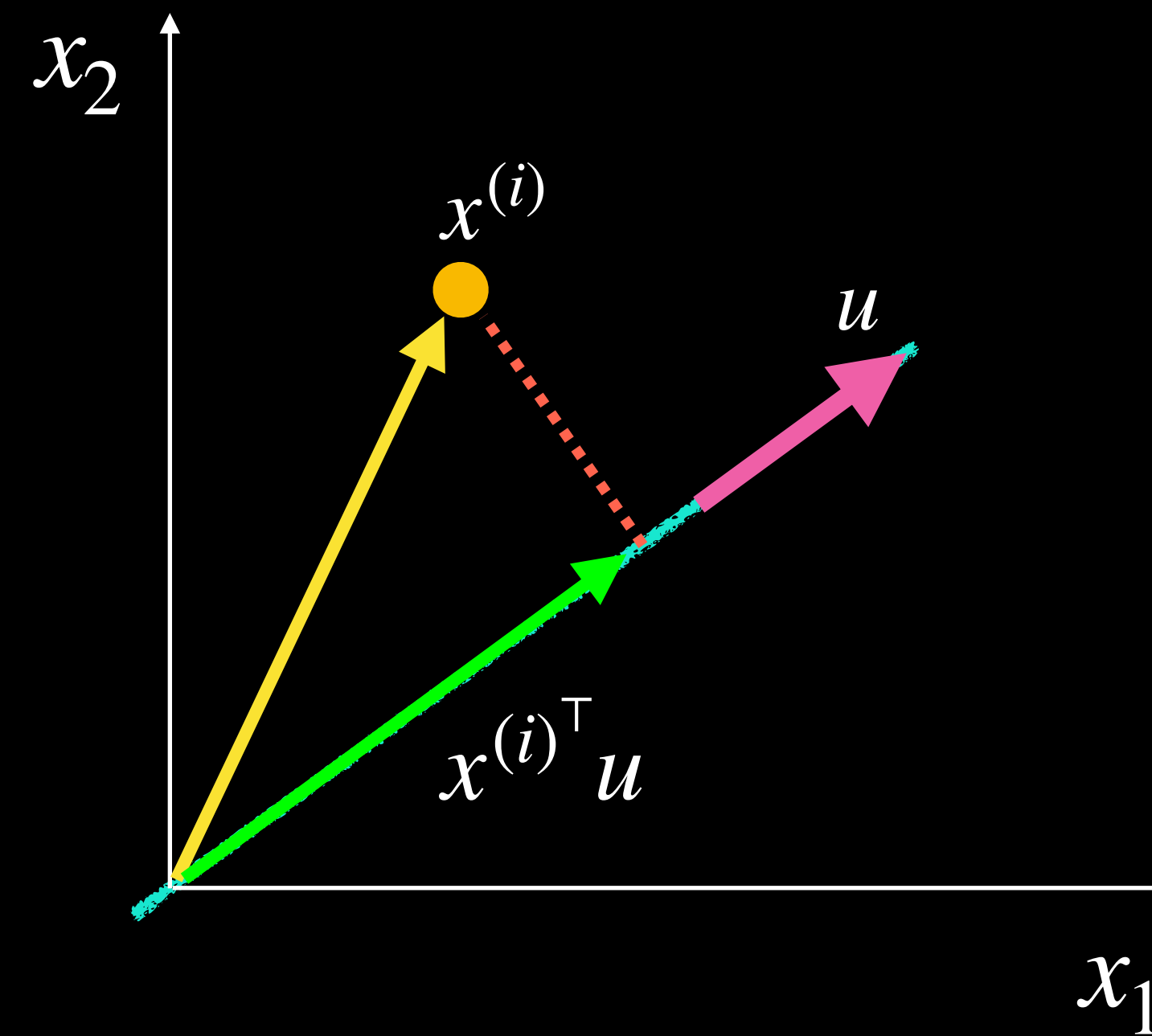


$$u_1, u_2, \dots, u_d \in \mathbb{R}^d$$

$$\lambda_1, \lambda_2, \dots, \lambda_d \in \mathbb{R}$$

```
# Compute the covariance matrix  
Sigma = X_centered.T @ X_centered
```

```
# Find eigenvalues and eigenvectors of the covariance matrix  
eigenvalues, eigenvectors = np.linalg.eig(Sigma)
```



Projection to k dimensions

$$y^{(i)} = \begin{bmatrix} u_1^\top x^{(i)} \\ u_2^\top x^{(i)} \\ \vdots \\ u_k^\top x^{(i)} \end{bmatrix} \in \mathbb{R}^k$$

Dataset

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Principal Component Analysis

To Maximize Variance of Projections

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$$\Sigma u = \lambda u$$

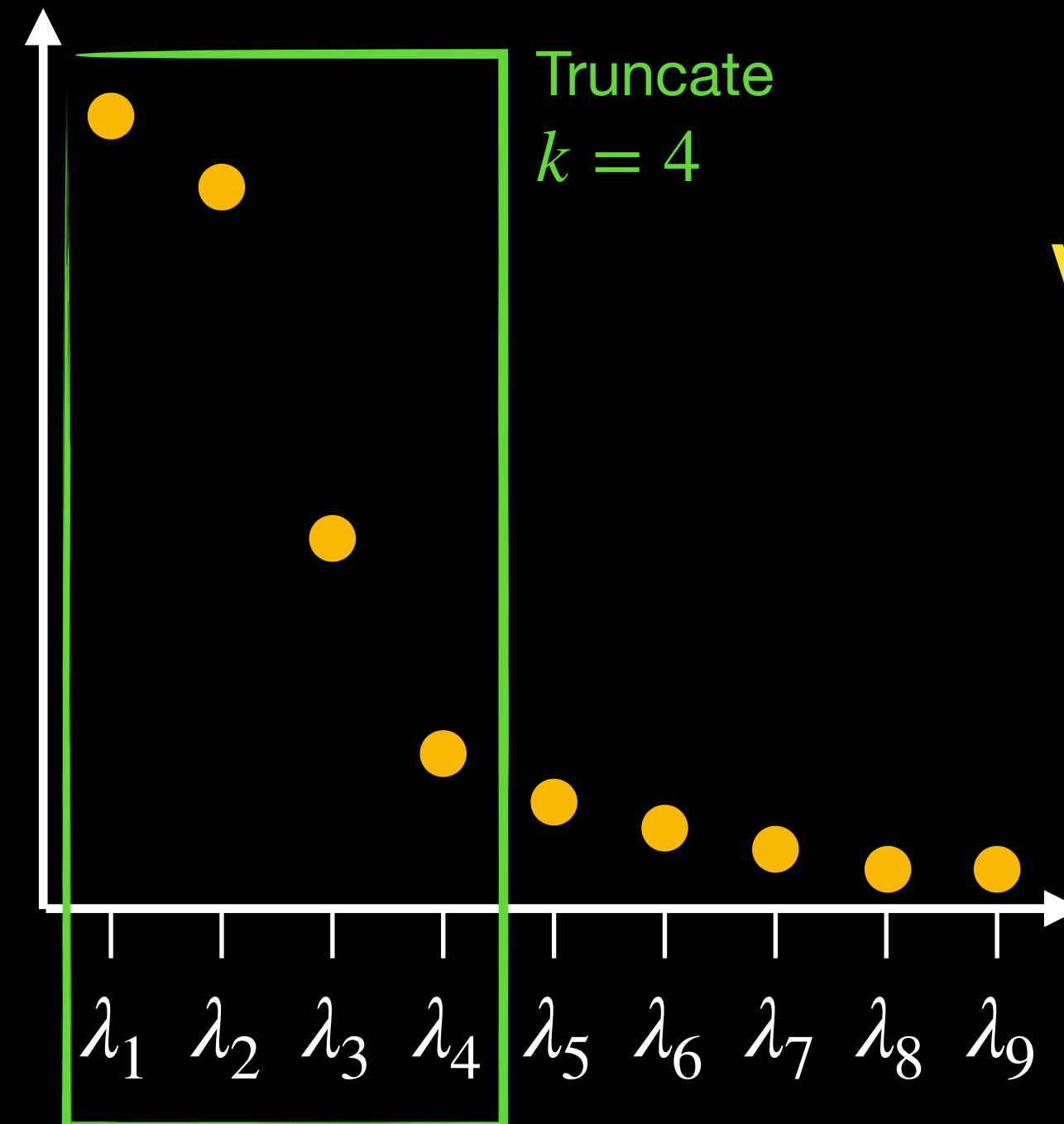
Eigenvalue problem

$$x \in \mathbb{R}^d$$
$$\Sigma \in \mathbb{R}^{d \times d}$$

Projection to k dimensions

$$y^{(i)} = \begin{bmatrix} u_1^\top x^{(i)} \\ u_2^\top x^{(i)} \\ \vdots \\ u_k^\top x^{(i)} \end{bmatrix} \in \mathbb{R}^k$$

Eigenvalue
Magnitude



Percentage of
Variance Preserved

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^d \lambda_i} \times 100$$

What are the dimensions of x , Σ and u in this case?

Principal Component Analysis

Typically, eigenvalues and eigenvectors

Solve

$$\Sigma v = \lambda v$$

naming eigenvectors v

More efficient

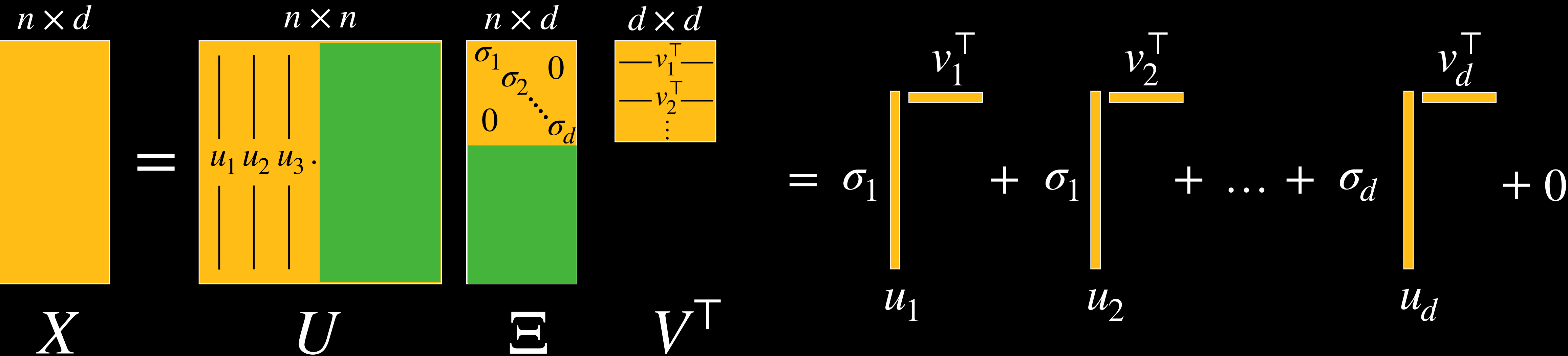
Solve

$$X = U \Sigma V^T$$

Singular Value Decomposition

U and V are unitary

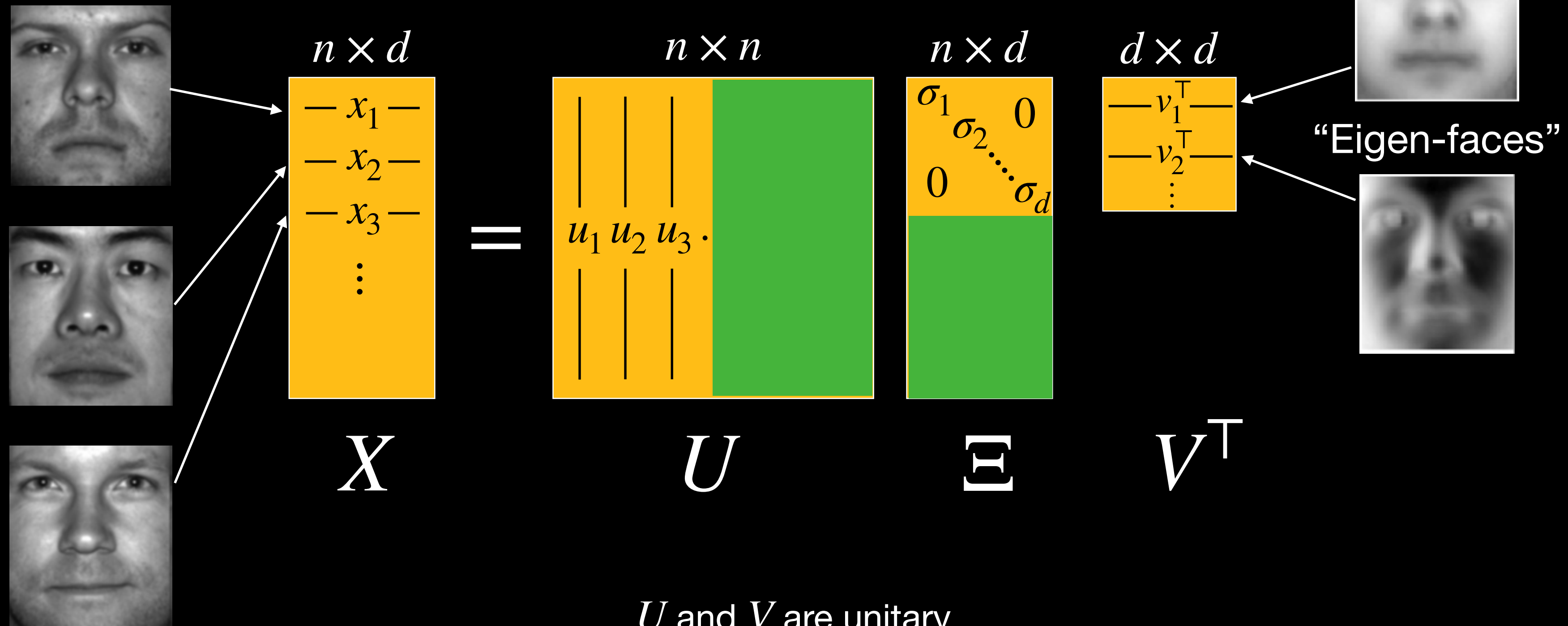
$$U^T U = U U^T = I$$
$$V^T V = V V^T = I$$



Singular Value Decomposition

$$X = U \Sigma V^T$$

A Dataset of Faces



U and V are unitary

$$U^T U = U U^T = I$$
$$V^T V = V V^T = I$$